

Lesson 2 Lesson-End Project

Designing Governance Solution Framework

Project Agenda: To design a solution for governance with respect to the given scenario

Description: You work as a cloud architect for a Fortune 500 organization.

You need to design a solution for developers that enables them to provide specific Azure Resources while adhering to the following requirements:

- Allow provisioning of VMs of a certain size only
- Permit provisioning of storage accounts and VMs in specific regions
- Require secure transfer to be enabled when creating a storage account

Tools required: Azure account with administrator access

Prerequisites: None

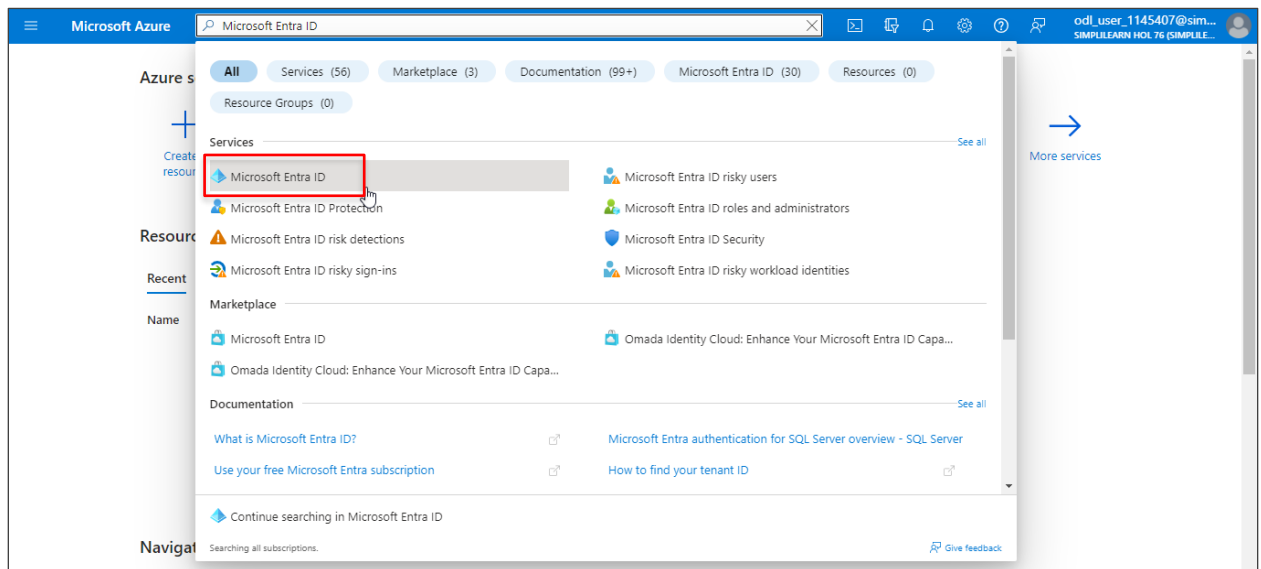
Expected deliverables: Illustrate the solution to the given scenario and allow all the policies as mentioned and add screenshots containing the respective outputs

Steps to be followed:

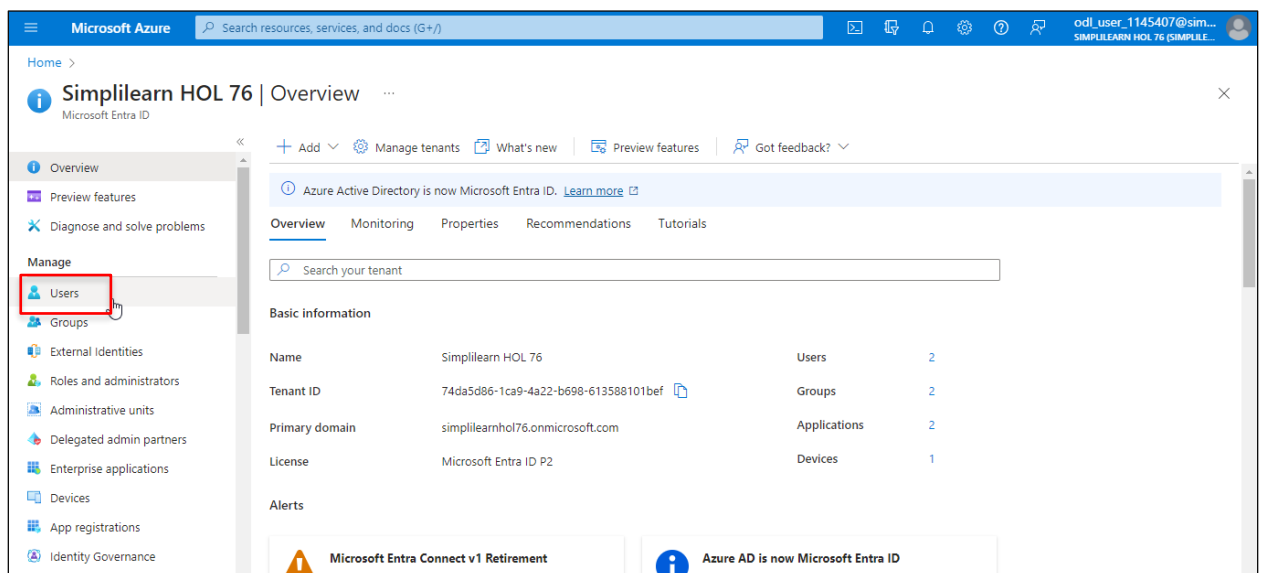
1. Create users
2. Create a group
3. Create a resource group to assign policy per the requirement
4. Assign a role to the created group at the resource group level
5. Assign the policies to the resource group

Step 1: Create users

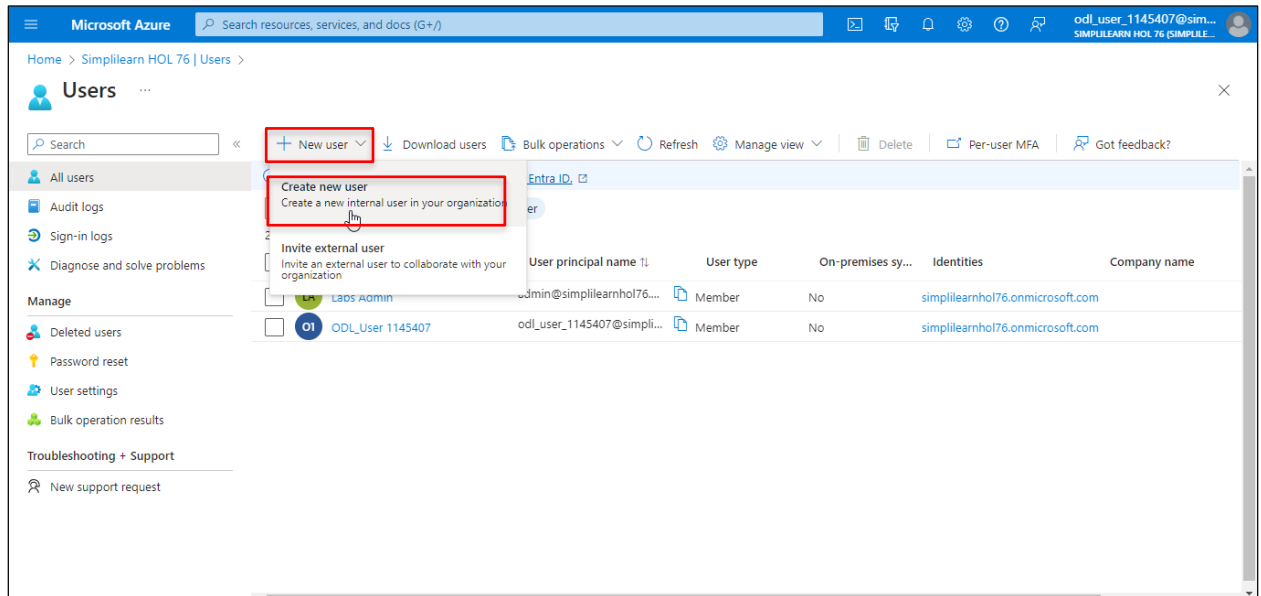
1.1 Log in to the Azure portal; search for and select **Microsoft Entra ID**



1.2 In the **Overview** page, under the **Manage** section, click on **Users**



1.3 Click on **+ New user** and then click on **Create new user** to create a new user account for one of your developers



1.4 In the **Create new user** page, fill in the required details as shown below and click on **Review + create** button:

The screenshot shows the 'Create new user' page in the Microsoft Azure portal. The 'Review + create' tab is selected. The form contains the following fields and options:

- User principal name:** demouser1 @ simpilearnhol76.onmicr... (Domain not listed)
- Mail nickname *:** demouser1 (Derive from user principal name is checked)
- Display name*:** demouser1
- Password *:** (Auto-generate password is checked)
- Account enabled ①:** (checked)

The 'Review + create' button is highlighted with a red box. Navigation buttons for '< Previous' and 'Next: Properties >' are also visible.

1.5 In the **Review + create** tab, click on **Create** after reviewing all the details

Microsoft Azure Search resources, services, and docs (G+)

Home > Simplilearn HOL 76 | Users > Users >

Create new user

Create a new internal user in your organization

Basics Properties Assignments **Review + create**

Basics

User principal name demouser1@simplilearnhol76.onmicrosoft.com

Display name demouser1

Mail nickname demouser1

Password

Account enabled Yes

Properties

User type Member

Assignments

Administrative units

Create < Previous Next >

After successful user creation, the following message will appear:

Microsoft Azure Search resources, services, and docs (G+)

Home > Simplilearn HOL 76 | Users >

Users

Search New user Download users Bulk operations Refresh Manage view Delete Per-user MFA Got feedback?

Azure Active Directory is becoming Microsoft Entra ID.

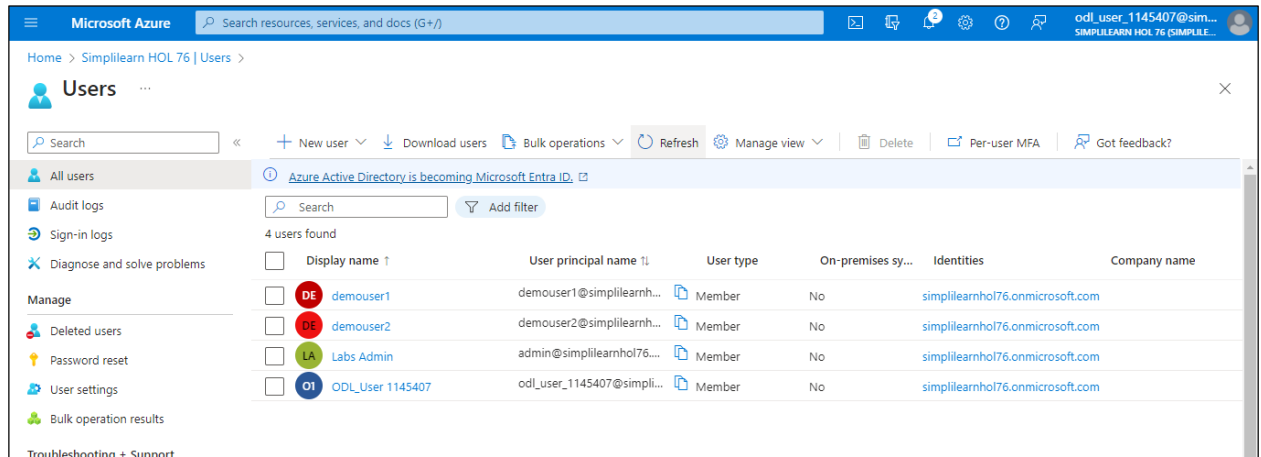
2 users found

Display name	User principal name	User type	On-premises sy...	Identities	Company name
LA Labs Admin	admin@simplilearnhol76....	Member	No	simplilearnhol76.onmicrosoft.com	
ODL_User 1145407	odl_user_1145407@simpli...	Member	No	simplilearnhol76.onmicrosoft.com	

Successfully created user
Successfully created user demouser1

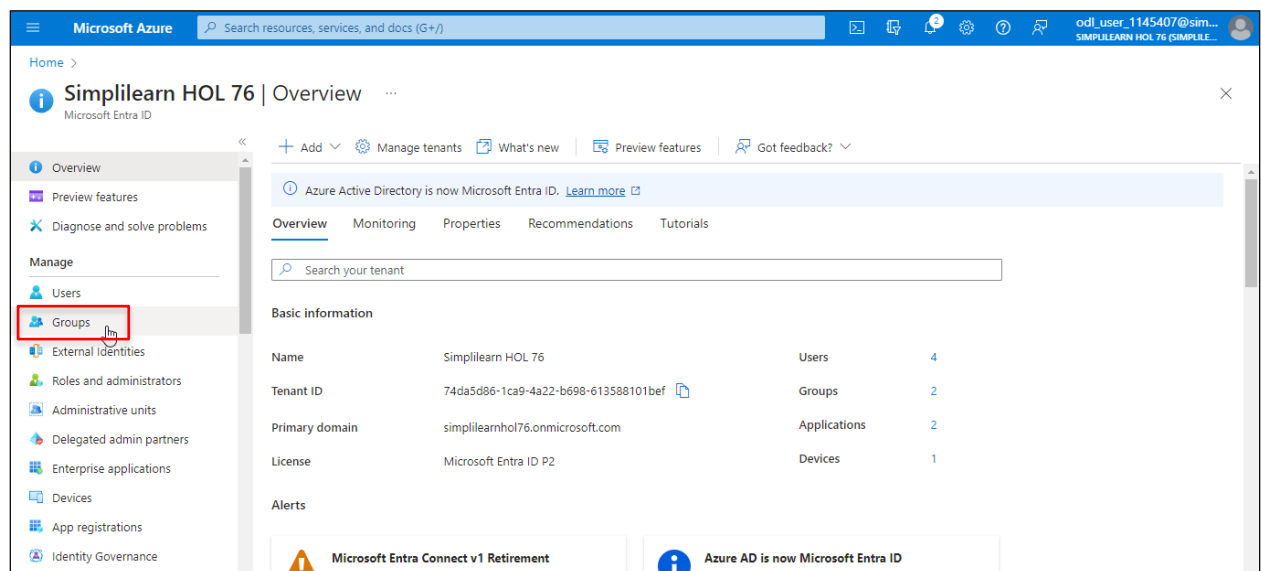
Note: Create one more user by following the same steps of user creation

As shown in the following screenshot, the two new users are created **demouser1** and **demouser2**.

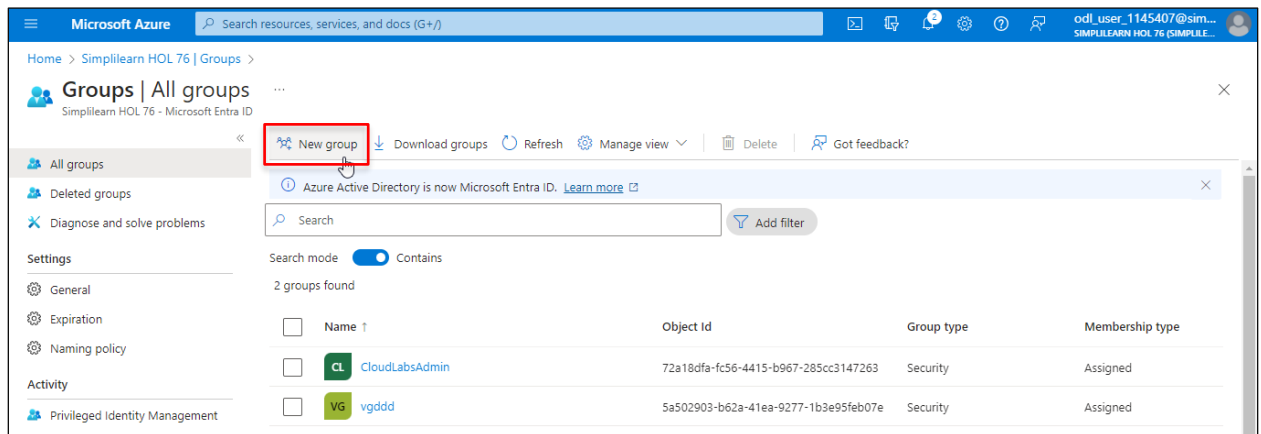


Step 2: Create a group

2.1 Navigate back to the **Overview** page of the **Microsoft Entra ID** and click on **Groups** under the **Manage** section



2.2 Click on **New Group** to create a new group for users



2.3 In the **New Group** page, fill in the required details and add the users to the group by clicking on the **No members selected**

The screenshot shows the 'New Group' page in the Microsoft Azure portal. The page contains several form fields and a 'Create' button at the bottom. A red box highlights the 'No members selected' link under the 'Members' section.

Group type * ⓘ
Security

Group name * ⓘ
Group1

Group description ⓘ
Policy Assign

Microsoft Entra roles can be assigned to the group ⓘ
Yes No

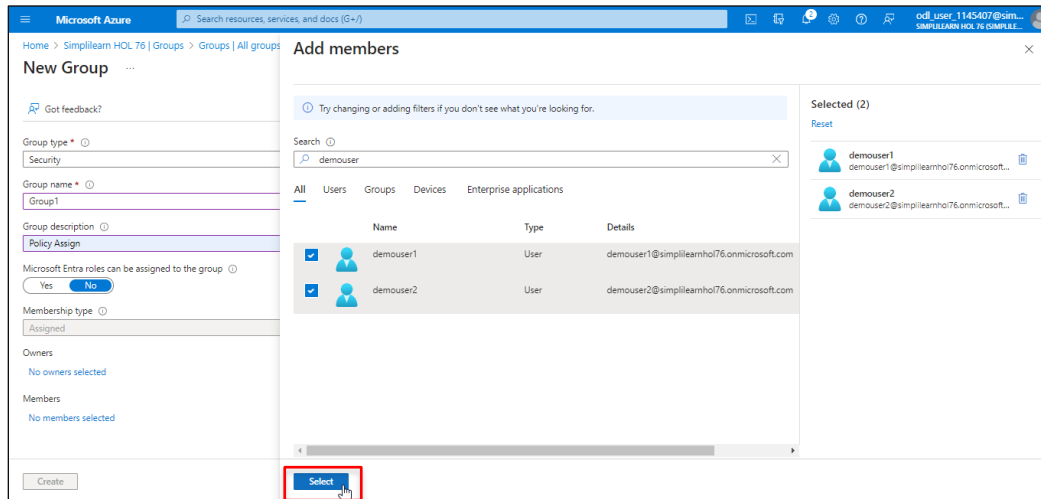
Membership type * ⓘ
Assigned

Owners
No owners selected

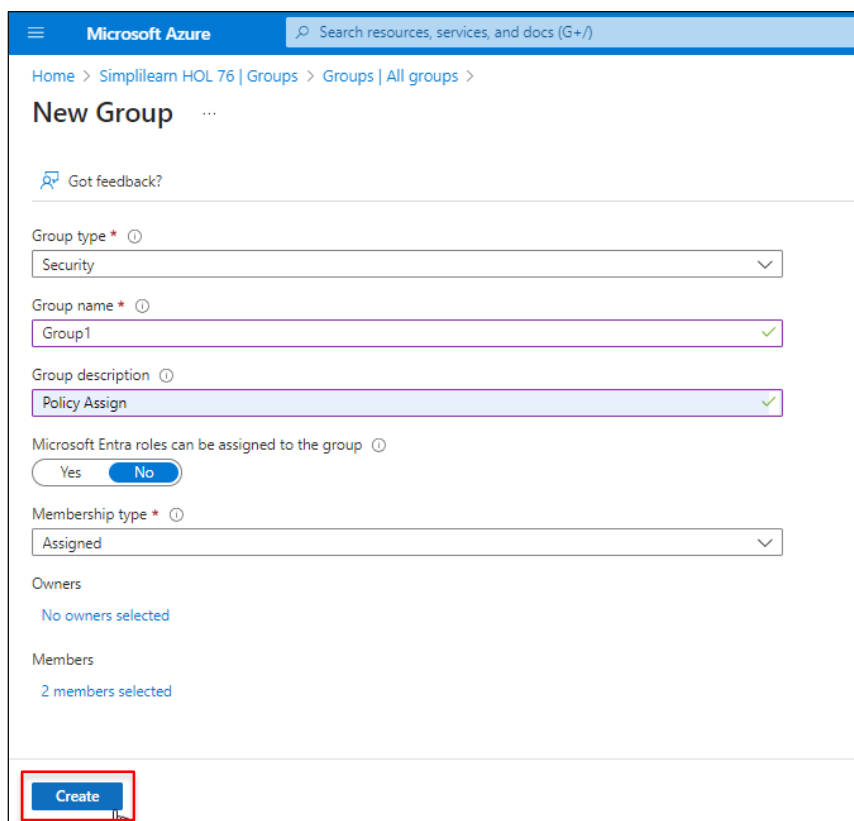
Members
No members selected

Create

2.4 In the **Add members** page, search for the users from Step 1, select the checkboxes next to the users, and then click on **Select**



2.5 Click on **Create**



After successful group creation, the following screen will appear:

Microsoft Azure | Groups | All groups

Search resources, services, and docs (G+ /)

odl_user_1145407@sim... SIMPLILEARN HOL 76 (SIMPLE...)

Home > Simplilearn HOL 76 | Groups >

Groups | All groups

Simplilearn HOL 76 - Microsoft Entra ID

New group Download groups Refresh Manage view Delete Got feedback?

Azure Active Directory is now Microsoft Entra ID. [Learn more](#)

Search

Add filter

Search mode Contains

2 groups found

Name	Object Id	Group type	Membership type	Email
CL CloudLabsAdmin	72a18dfa-fc56-4415-b967-285cc3147263	Security	Assigned	
VG vgddd	5a502903-b62a-41ea-9277-1b3e95feb07e	Security	Assigned	

Settings: General, Expiration, Naming policy, Activity, Privileged Identity Management, Access reviews

Note: If the newly created group is not visible, then click on **Refresh**

Microsoft Azure | Groups | All groups

Search resources, services, and docs (G+ /)

odl_user_1145407@sim... SIMPLILEARN HOL 76 (SIMPLE...)

Home > Simplilearn HOL 76 | Groups >

Groups | All groups

Simplilearn HOL 76 - Microsoft Entra ID

New group Download groups Refresh Manage view Delete Got feedback?

Azure Active Directory is now Microsoft Entra ID. [Learn more](#)

Search

Add filter

Search mode Contains

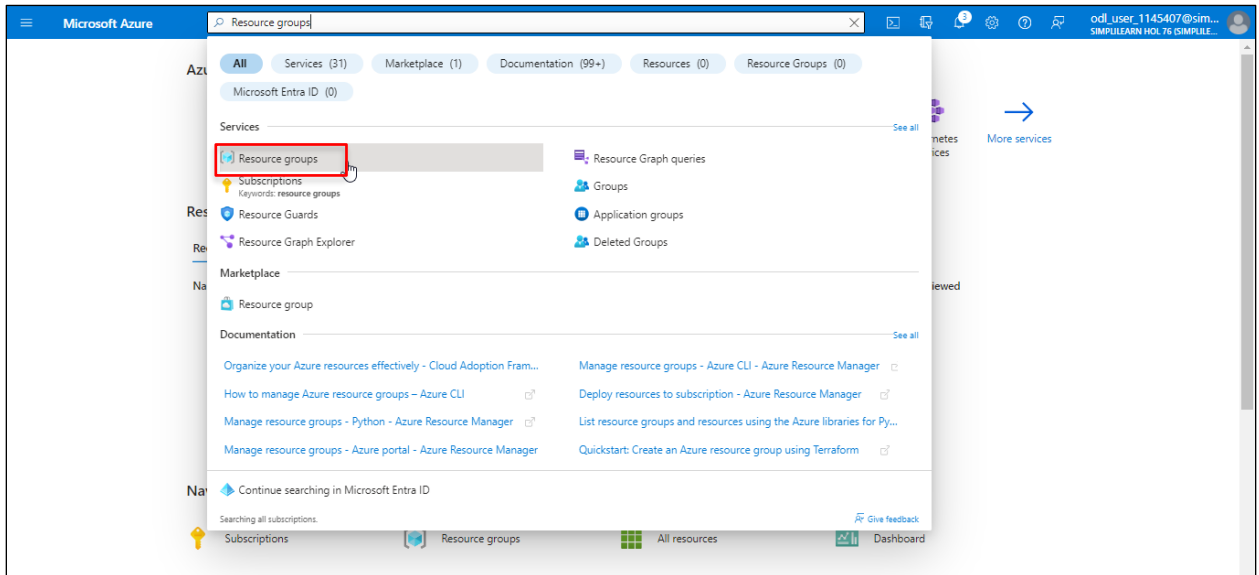
3 groups found

Name	Object Id	Group type	Membership type	Email
CL CloudLabsAdmin	72a18dfa-fc56-4415-b967-285cc3147263	Security	Assigned	
GR Group1	cbe60960-a506-4def-9872-ab0b8ce6f7c	Security	Assigned	
VG vgddd	5a502903-b62a-41ea-9277-1b3e95feb07e	Security	Assigned	

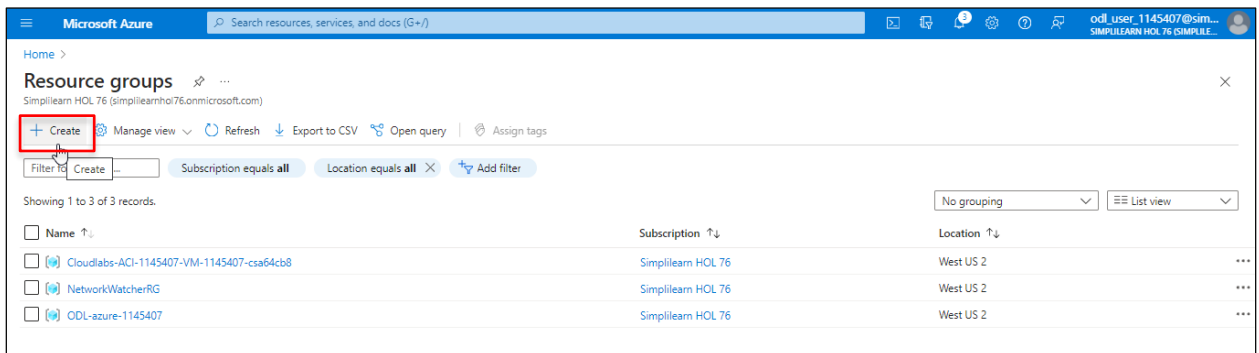
Settings: General, Expiration, Naming policy, Activity, Privileged Identity Management, Access reviews, Audit logs, Bulk operation results

Step 3: Create a resource group to assign policy per the requirement

3.1 Search for and select **Resource groups** in the search box



3.2 In the **Resource groups** page, click on + Create



3.3 In the **Create a resource group** page, enter the required details and click on **Review + create**

The screenshot shows the Microsoft Azure portal interface for creating a resource group. The top navigation bar includes the Microsoft Azure logo and a search bar. The breadcrumb trail indicates the user is in the 'Resource groups' section. The main heading is 'Create a resource group'. Below this, there are three tabs: 'Basics', 'Tags', and 'Review + create'. The 'Basics' tab is currently selected. A descriptive paragraph explains that a resource group is a container for related resources. Below this, the 'Project details' section contains two dropdown menus: 'Subscription' (set to 'Simplilearn HOL 76') and 'Resource group' (set to 'RGPA' with a green checkmark). The 'Resource details' section contains a 'Region' dropdown menu set to '(US) East US'. At the bottom of the form, there are three buttons: 'Review + create' (highlighted with a red box and a mouse cursor), '< Previous', and 'Next : Tags >'.

Microsoft Azure

Search resources, services, and docs (G+/)

Home > Resource groups >

Create a resource group

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Project details

Subscription * ⓘ Simplilearn HOL 76

Resource group * ⓘ RGPA

Resource details

Region * ⓘ (US) East US

Review + create < Previous Next : Tags >

3.4 After the validation is passed, click on **Create**

Microsoft Azure

Search resources, services, and docs (G+)

Home > Resource groups >

Create a resource group

✓ Validation passed.

Basics Tags Review + create

Basics

Subscription: Simplilearn HOL 76

Resource group: RGPA

Region: East US

Tags

None

Create < Previous Next > Download a template for automation

After successful resource group creation, the following screen will appear:

Microsoft Azure

Search resources, services, and docs (G+)

Home >

Resource groups

Simplilearn HOL 76 (simplilearnhol76.onmicrosoft.com)

+ Create Manage view Refresh Export to CSV Open query Assign tags

Filter for any field... Subscription equals all Location equals all Add filter

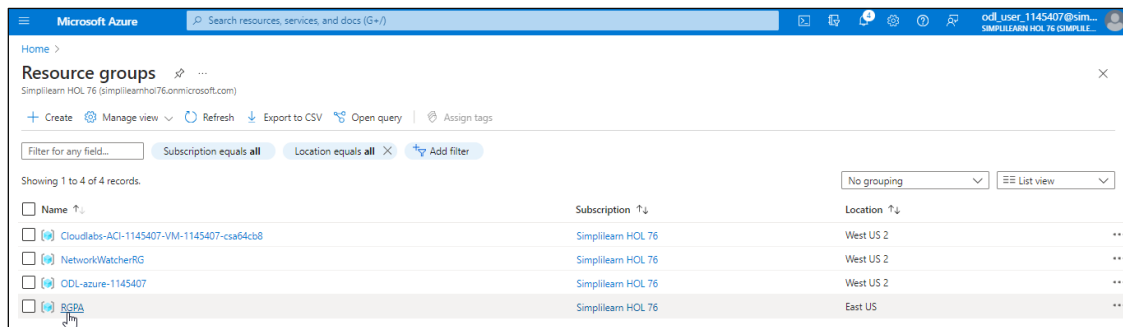
Showing 1 to 4 of 4 records.

Name ↑↓	Subscription ↑↓	Location ↑↓
Cloudlabs-ACI-1145407-VM-1145407-csa64cb8	Simplilearn HOL 76	West US 2
NetworkWatcherRG	Simplilearn HOL 76	West US 2
ODL-azure-1145407	Simplilearn HOL 76	West US 2
RGPA	Simplilearn HOL 76	East US

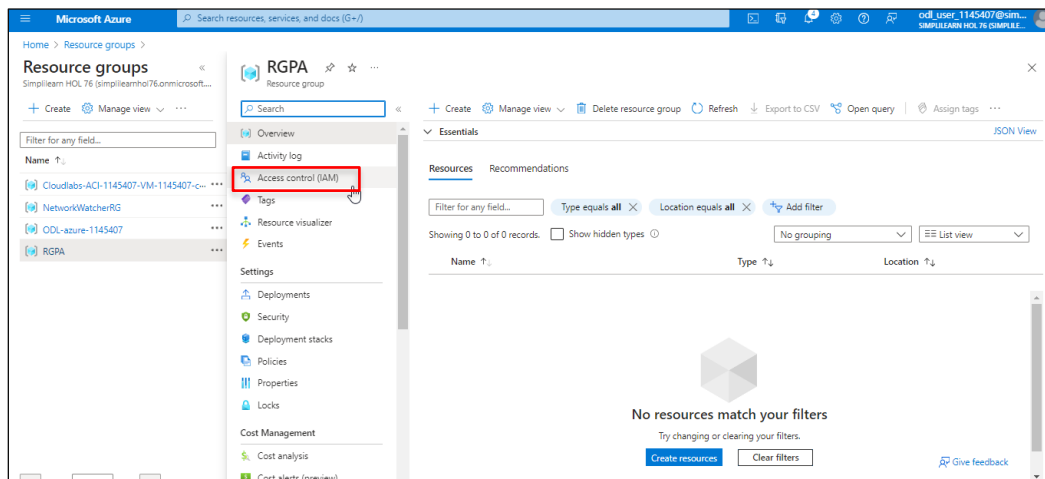
Resource group created
Creating resource group 'RGPA' in subscription 'Simplilearn HOL 76' succeeded.
Go to resource gr... Pin to dashbo...

Step 4: Assign a role to the created group at the resource group level

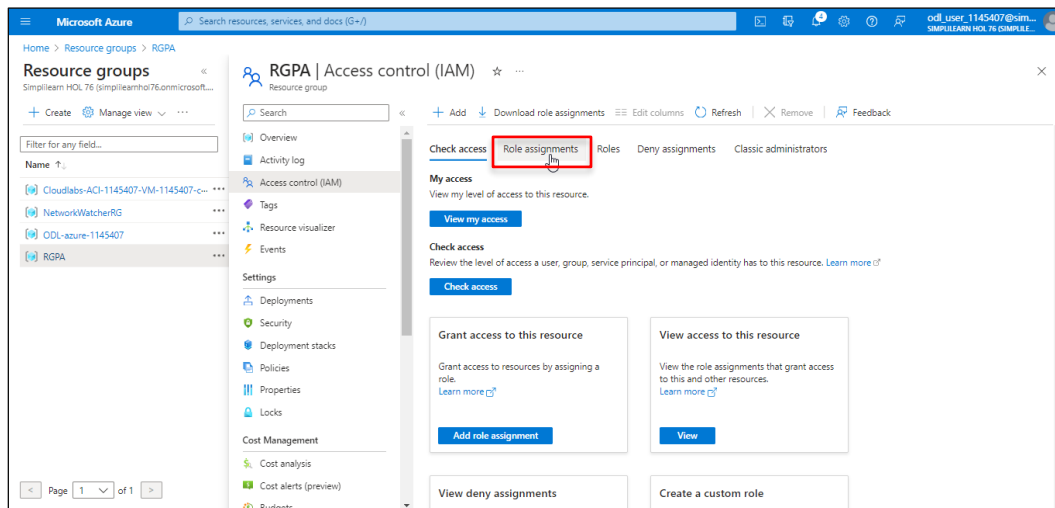
4.1 Click on the resource group **RGPA** created in the previous step



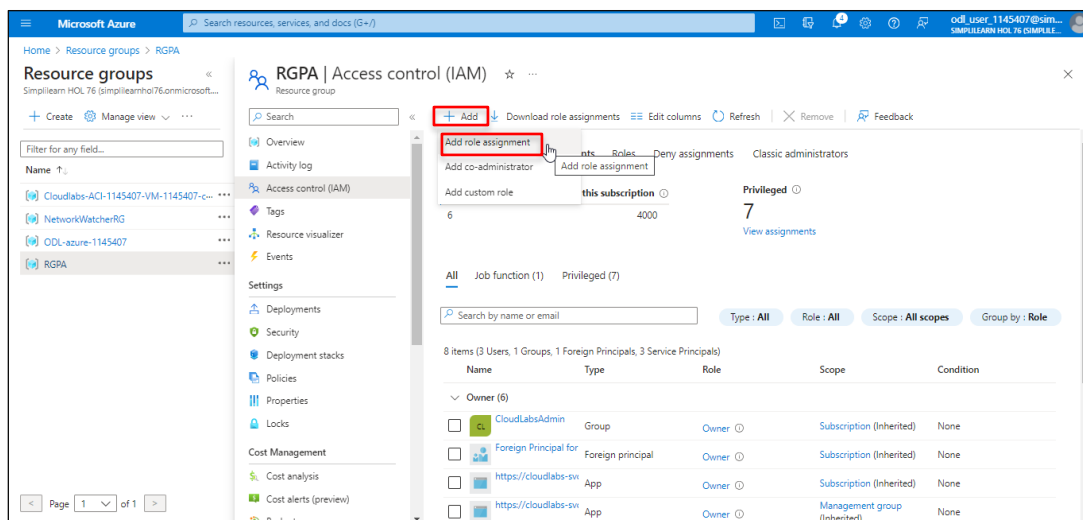
4.2 Click on **Access control (IAM)**



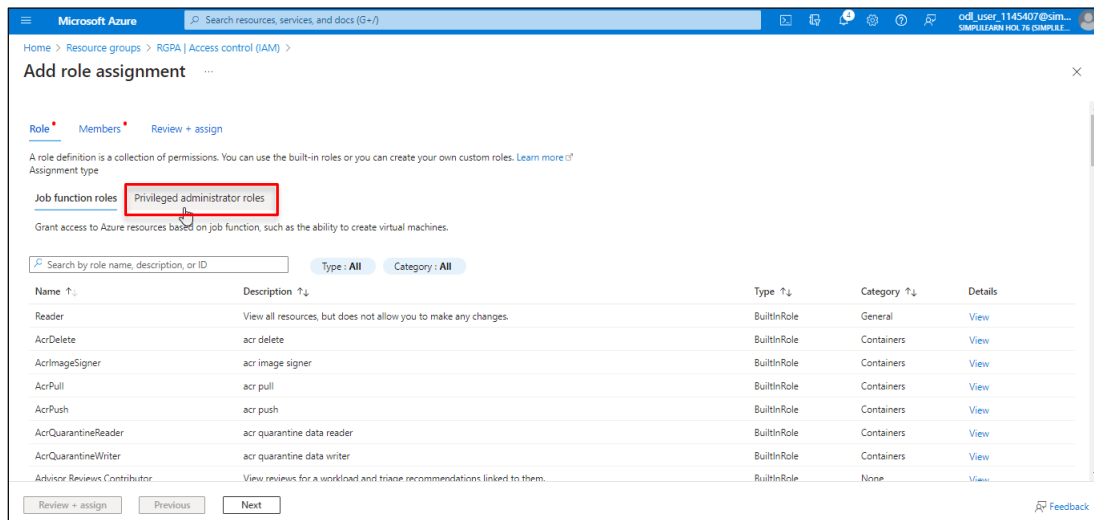
4.3 On the **Access control (IAM)** page, click on **Role assignments** tab



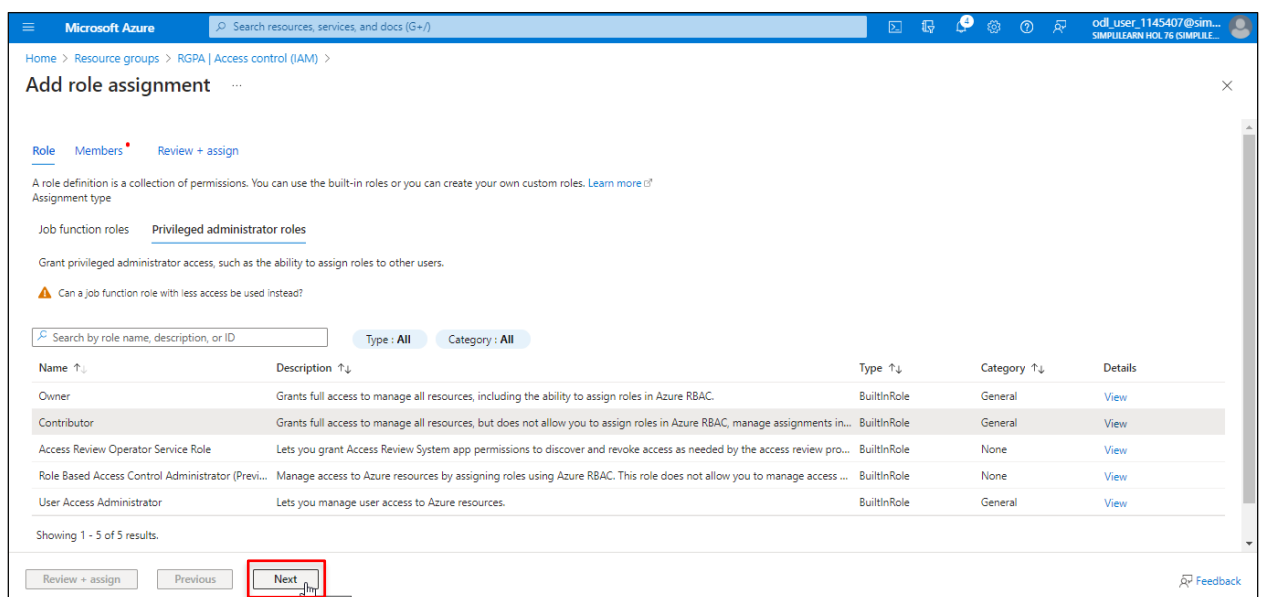
4.4 Click on **+ Add** and then click on **Add role assignment** to assign the role to the developer group



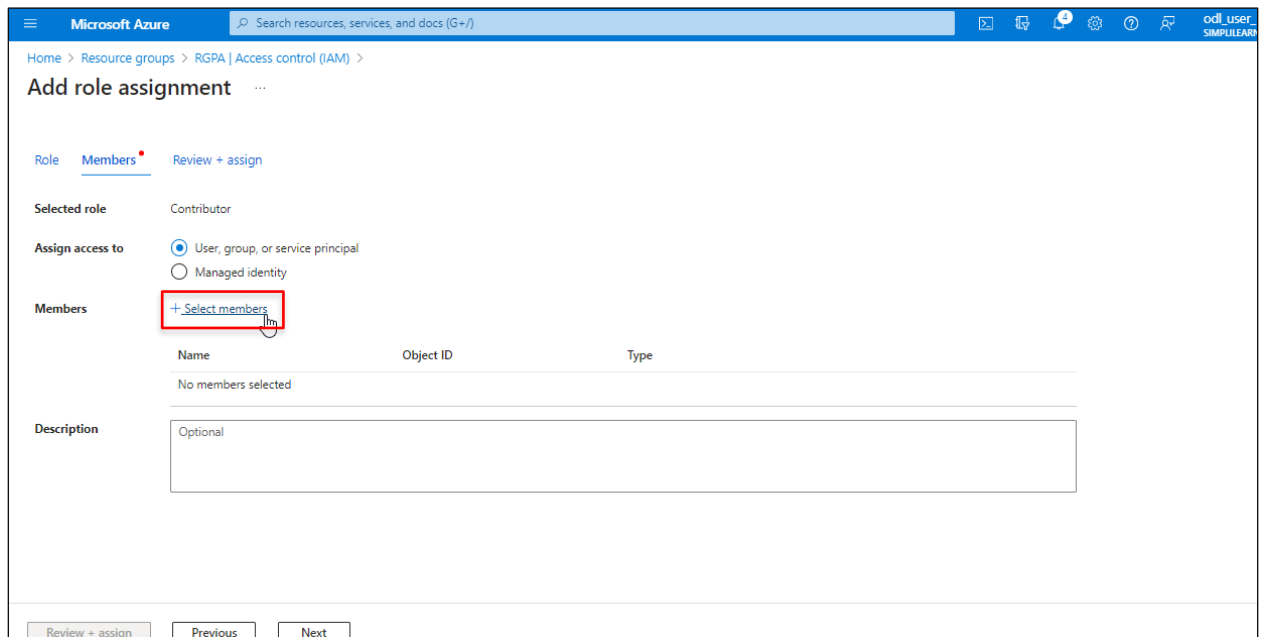
4.5 In the **Add role assignment** page, click on **Privileged administrator roles** tab



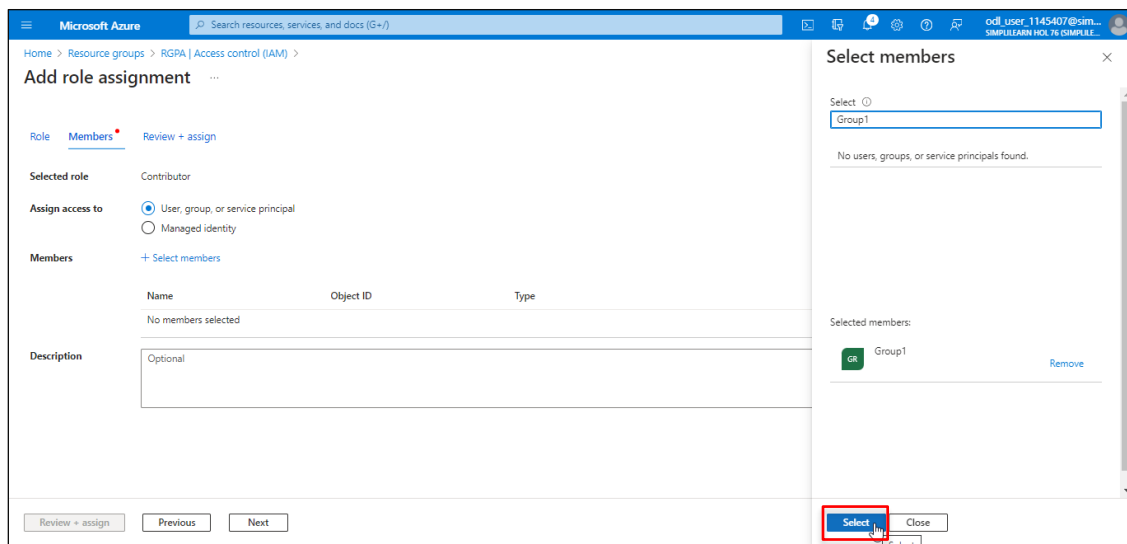
4.6 Click on the **Contributor** under **Name**, to assign a contributor role to the developer group. Then, click on **Next**.



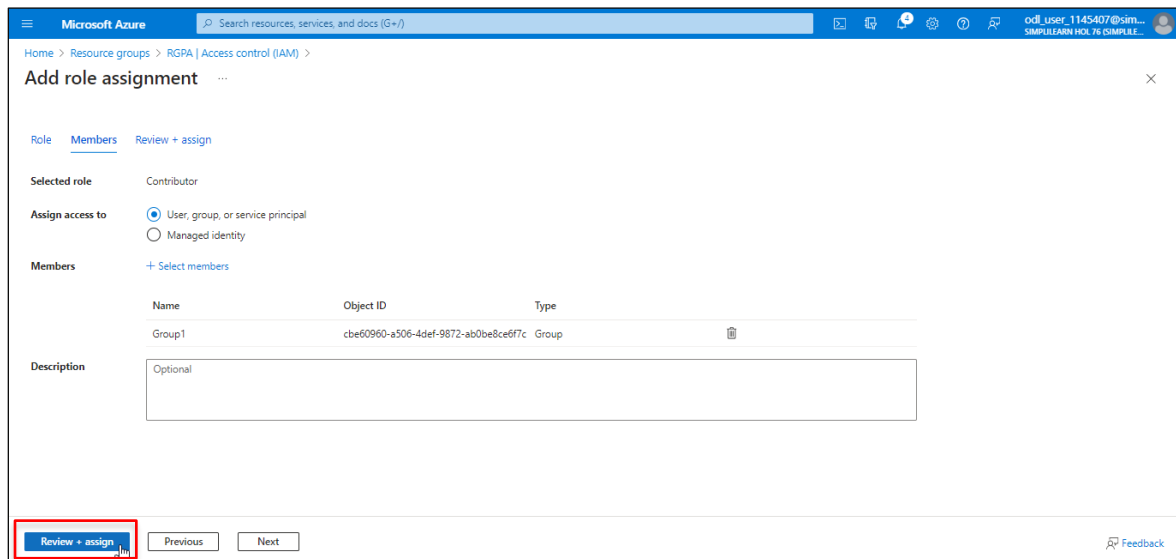
4.7 In the **Members** tab, click on **+ Select members**



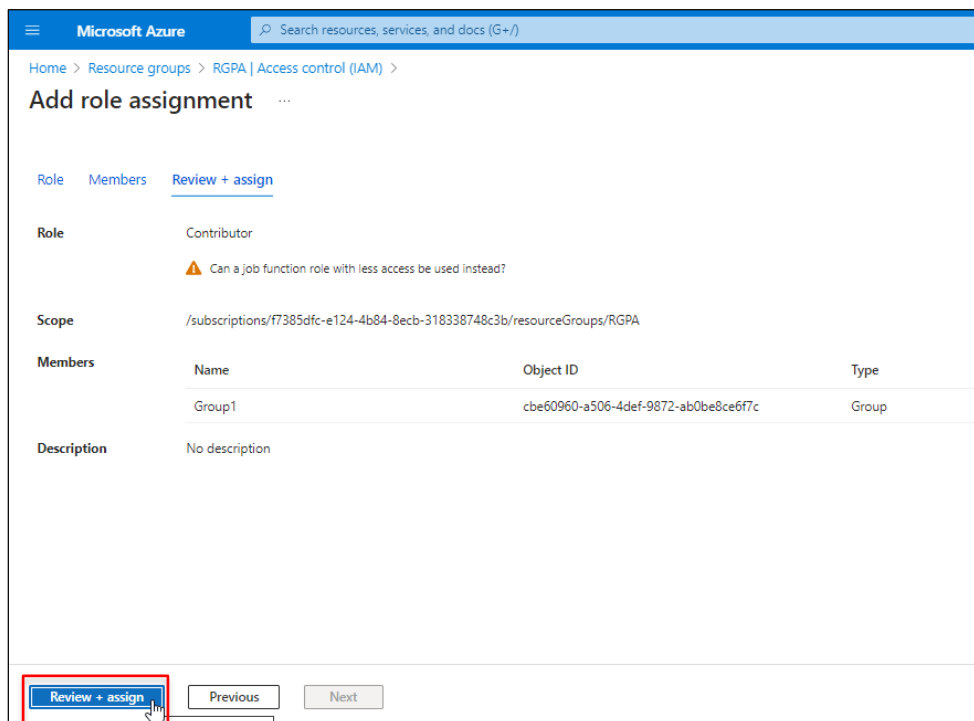
4.8 In the **Select members** page, search for and select the group created in step 2 (**Group1**), then click on **Select**



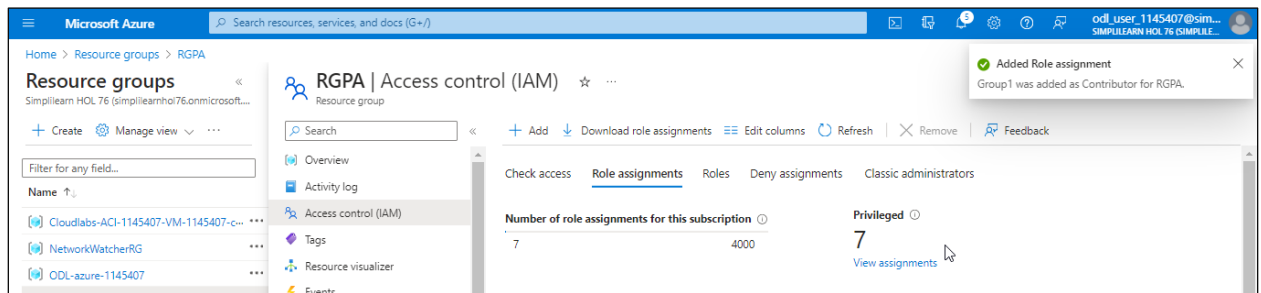
4.9 In the **Members** tab, click on **Review + assign** button



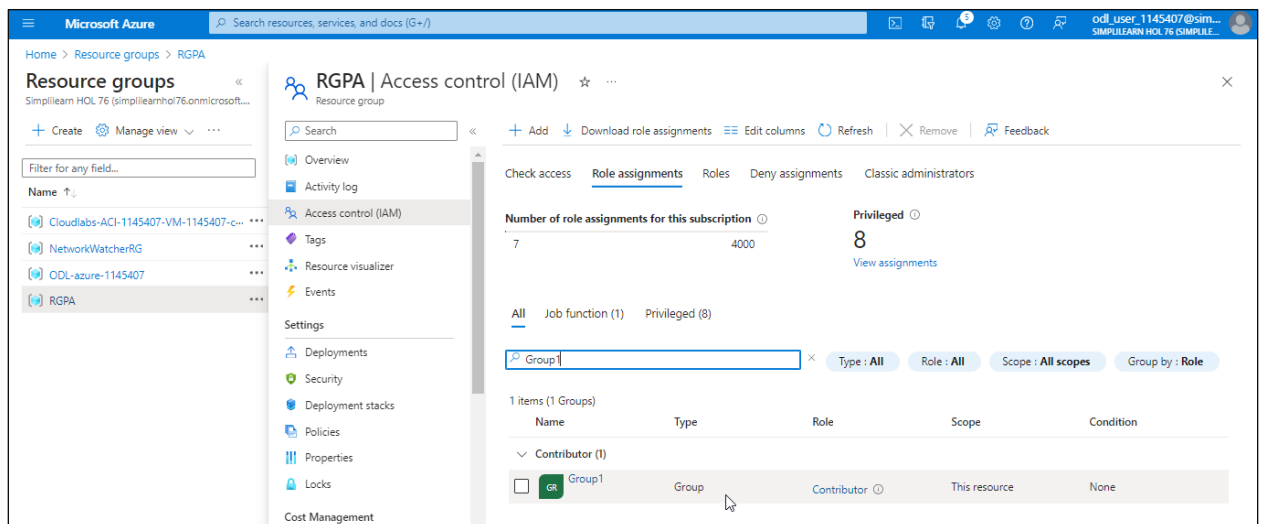
4.10 In the **Review + assign** tab, click on **Review + assign** button



After a successful role assignment, the following message will appear:

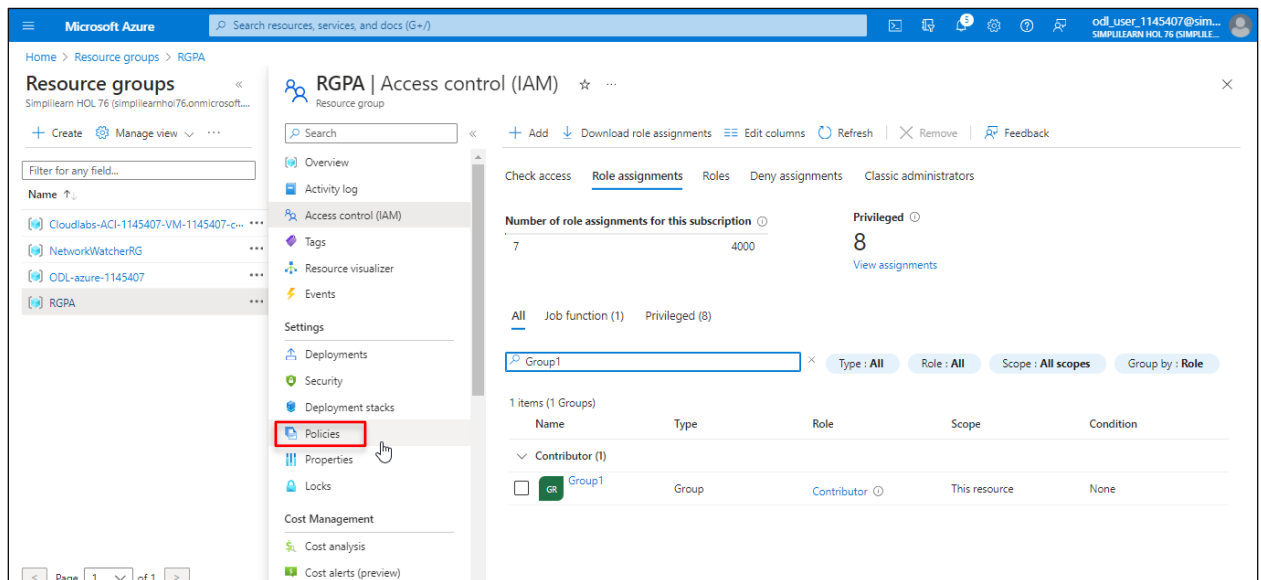


As shown in the following screenshot, **Contributor** role has been assigned to the **Group1**:

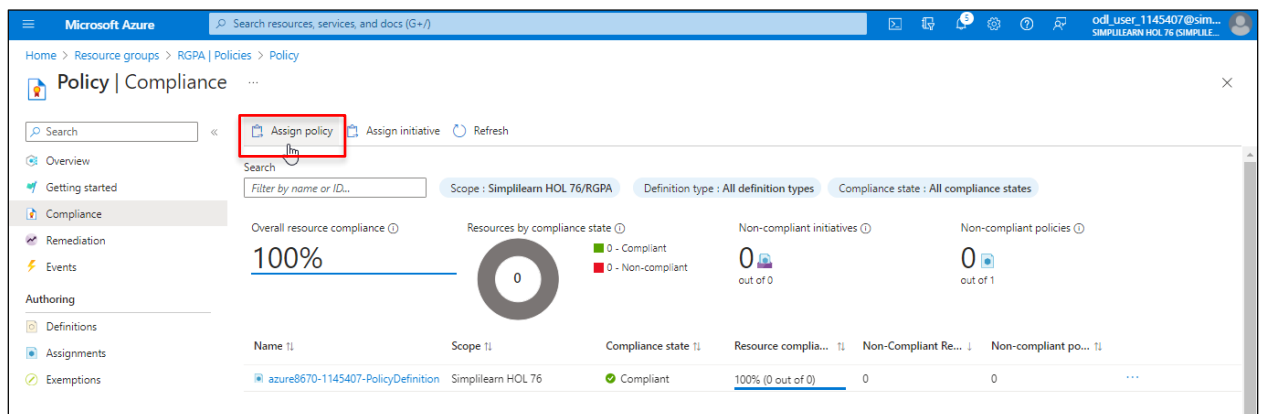


Step 5: Assign the policies to the resource group

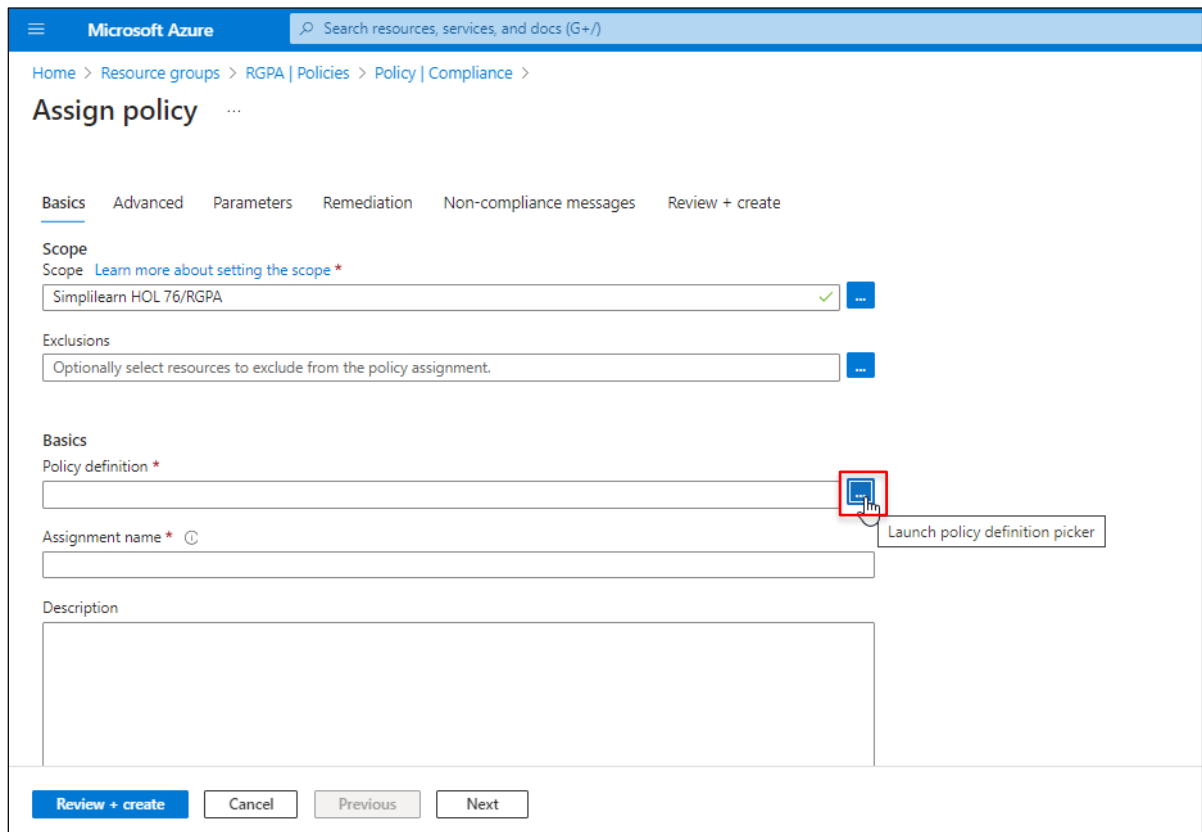
5.1 In the Access control (IAM) page, click on Policies under Settings



5.2 Click on Assign Policy



5.3 In the **Assign policy** page, select policies per the given requirements and set the parameters by clicking on the **Policy definition** option



Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups > RGPA | Policies > Policy | Compliance >

Assign policy ...

Basics Advanced Parameters Remediation Non-compliance messages Review + create

Scope
 Scope [Learn more about setting the scope *](#)
 Simplilearn HOL 76/RGPA ✓ ...

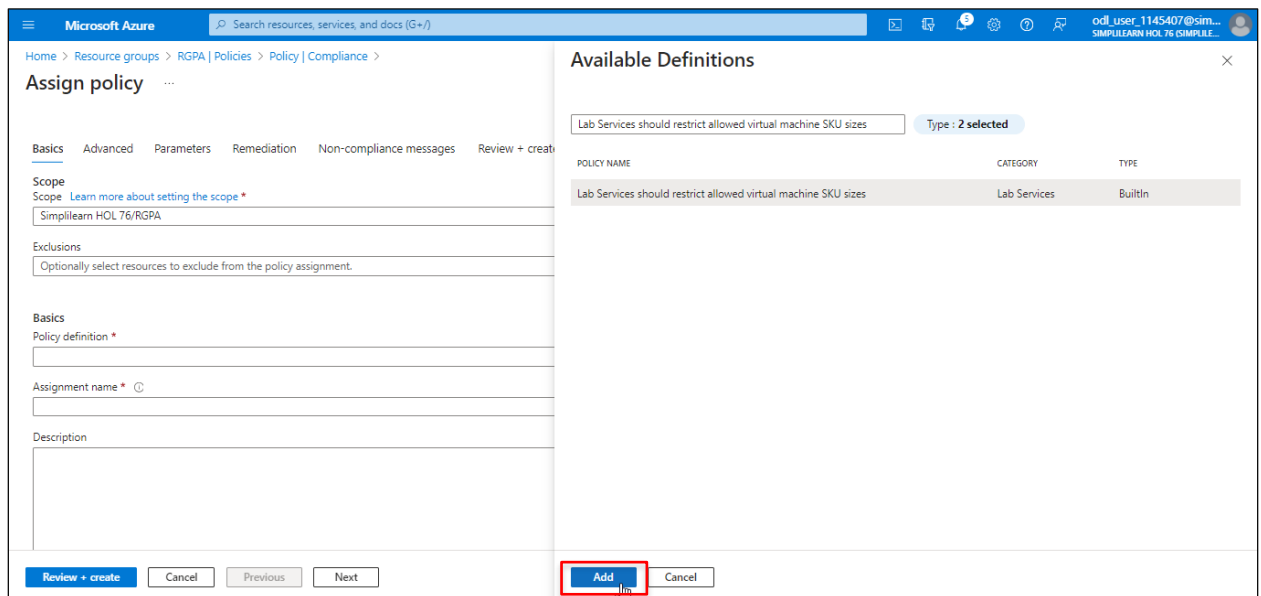
Exclusions
 Optionally select resources to exclude from the policy assignment. ...

Basics
 Policy definition *
 ...
 Assignment name * ⓘ
 ...
 Description
 ...

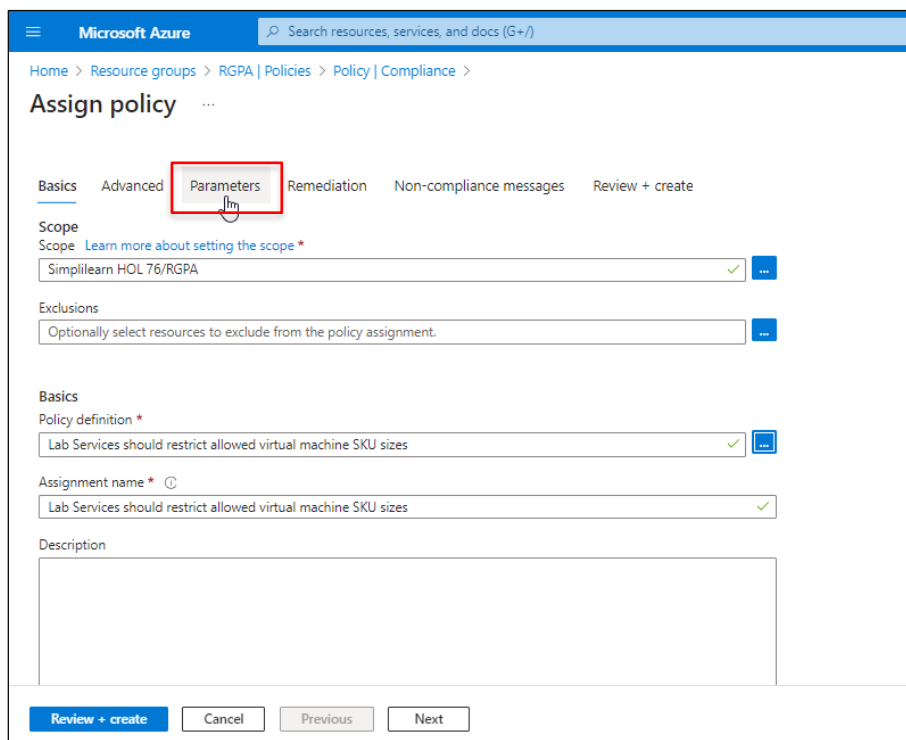
Launch policy definition picker

Review + create Cancel Previous Next

5.4 In the **Available Definitions** page, select a policy that allows only certain virtual machine size SKUs by selecting **Lab Services should restrict allowed virtual machine SKU sizes**, then click on **Add**



5.5 Click on **Parameters** tab



5.6 In the **Parameters** tab, select few **Allowed SKU names** by clicking on the dropdown and making selections as shown in the screenshot below:

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups > RGPA | Policies > Policy | Compliance >

Assign policy ...

Basics Advanced **Parameters** Remediation Non-compliance messages Review + create

Search by parameter name ☒ Only show parameters that need input or review

Allowed SKU names * ⓘ

16 selected

☐ Select all

- ☒ Standard_Fsv2_2_4GB_64_S_SSD
- ☒ Standard_Fsv2_2_4GB_128_S_SSD
- ☒ Classic_Fsv2_2_4GB_128_S_SSD
- ☐ Basic
- ☒ Classic_Fsv2_4_8GB_128_S_SSD
- ☐ Standard
- ☒ Classic_Fsv2_8_16GB_128_S_SSD
- ☐ Large
- ☒ Standard_Dsv4_2_8GB_128_S_SSD

Review + create Cancel Previous Next

5.7 Click on **Review + create**

The screenshot shows the 'Assign policy' page in the Microsoft Azure portal. The breadcrumb navigation is 'Home > Resource groups > RGPA | Policies > Policy | Compliance >'. The page title is 'Assign policy'. There are five tabs: 'Basics', 'Advanced', 'Parameters' (which is selected), 'Remediation', and 'Non-compliance messages'. Below the tabs, there is a search bar labeled 'Search by parameter name' and a checkbox labeled 'Only show parameters that need input or review' which is checked. Below that is a dropdown menu labeled 'Allowed SKU names *' with a help icon, showing '16 selected'. At the bottom of the page, there are four buttons: 'Review + create' (highlighted with a mouse cursor), 'Cancel', 'Previous', and 'Next'.

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups > RGPA | Policies > Policy | Compliance >

Assign policy

Basics Advanced **Parameters** Remediation Non-compliance messages Review + create

Search by parameter name ☒ Only show parameters that need input or review

Allowed SKU names * ⓘ
16 selected

Review + create Cancel Previous Next

5.8 In the **Review + create** tab, click on **Create**

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups > RGPA | Policies > Policy | Compliance >

Assign policy

Basics Advanced Parameters Remediation Non-compliance messages Review + create

Basics

Scope	Simplilearn HOL 76/RGPA
Exclusions	--
Policy definition	Lab Services should restrict allowed virtual machine SKU sizes
Assignment name	Lab Services should restrict allowed virtual machine SKU sizes
Description	--
Policy enforcement	Enabled
Assigned by	ODL_User 1145407

Advanced

Resource selectors (preview)

No selectors associated with this assignment.

Overrides (preview)

No overrides associated with this assignment.

Create Cancel Previous Next

The following message will appear on the screen once the policy assignment is successful:

Microsoft Azure Search resources, services, and docs (G+)

odl_user_1145407@sim... SIMPLILEARN HOL 76 SIMPLILEARN

Home > Resource groups > RGPA | Policies > Policy

Policy | Compliance

Search Filter by name or ID... Assign policy Assign initiative Refresh

Overview demoDB_1
Getting started testcosmosdb1
Compliance
Remediation
Events

Authoring
Definitions
Assignments

Overall resource compliance 100%

Resources by compliance state 0
0 - Compliant
0 - Non-compliant

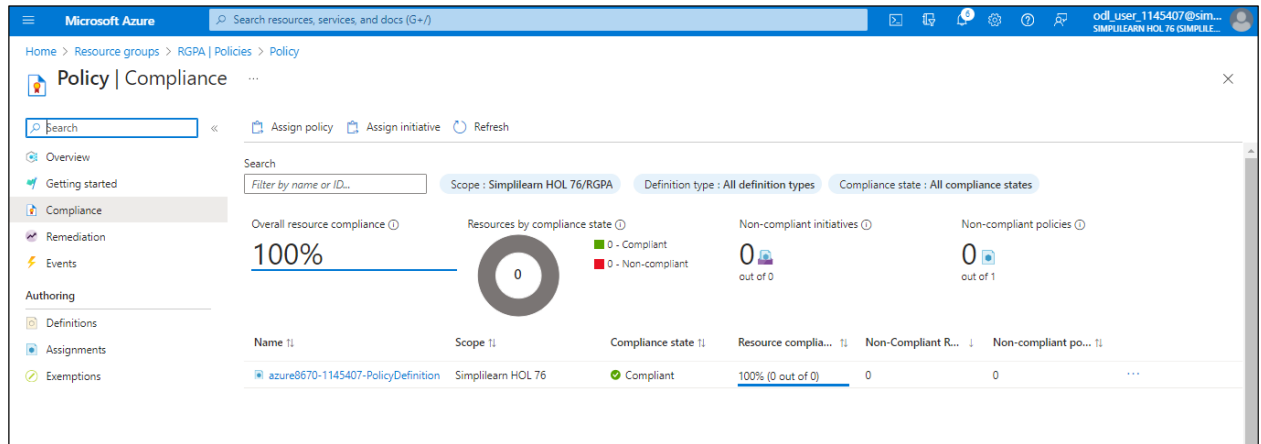
Non-compliant initiatives 0 out of 0

Non-compliant policies 0 out of 1

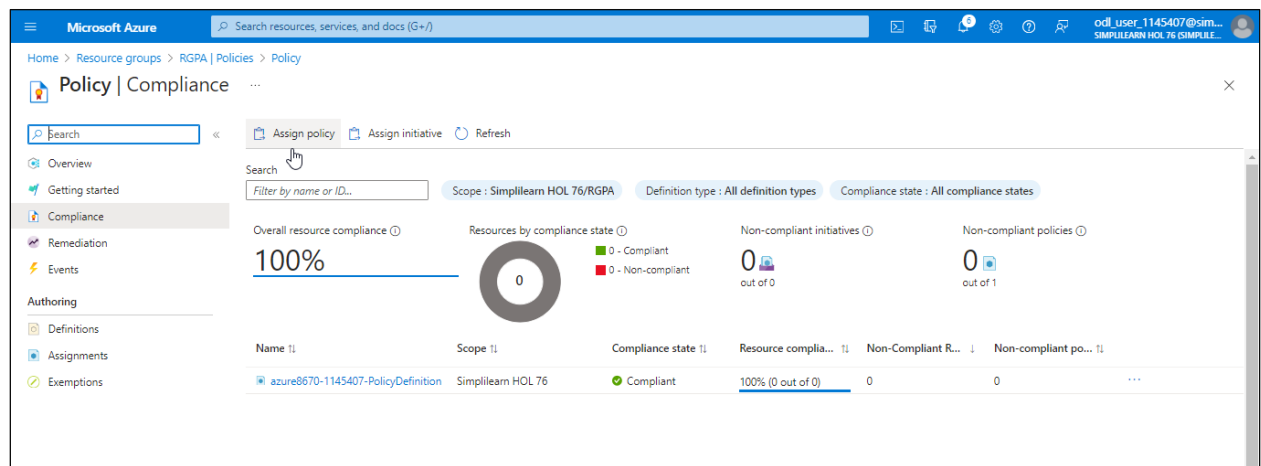
Creating policy assignment succeeded
Creating policy assignment 'Lab Services should restrict allowed virtual machine SKU sizes' in 'Simplilearn HOL 76/RGPA' was successful. Please note that the assignment takes around 5-15 minutes to take effect.

Name	Scope	Compliance state	Resource compliance	Non-Compliant Resources	Non-compliant policies

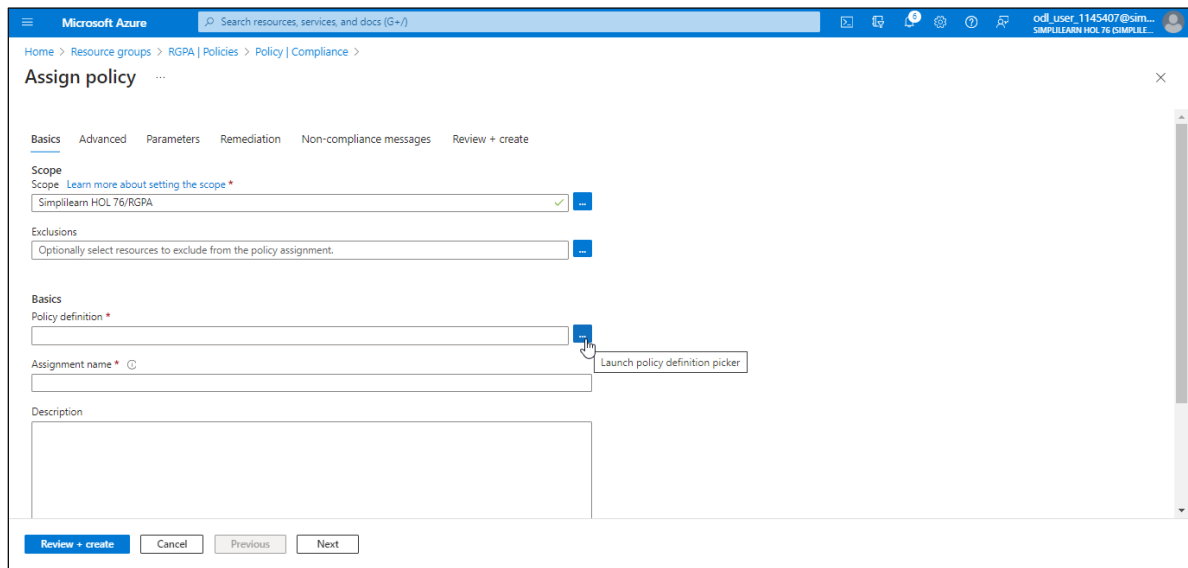
On the **Policy | Compliance** page, the first policy is added as shown in the screenshot below:



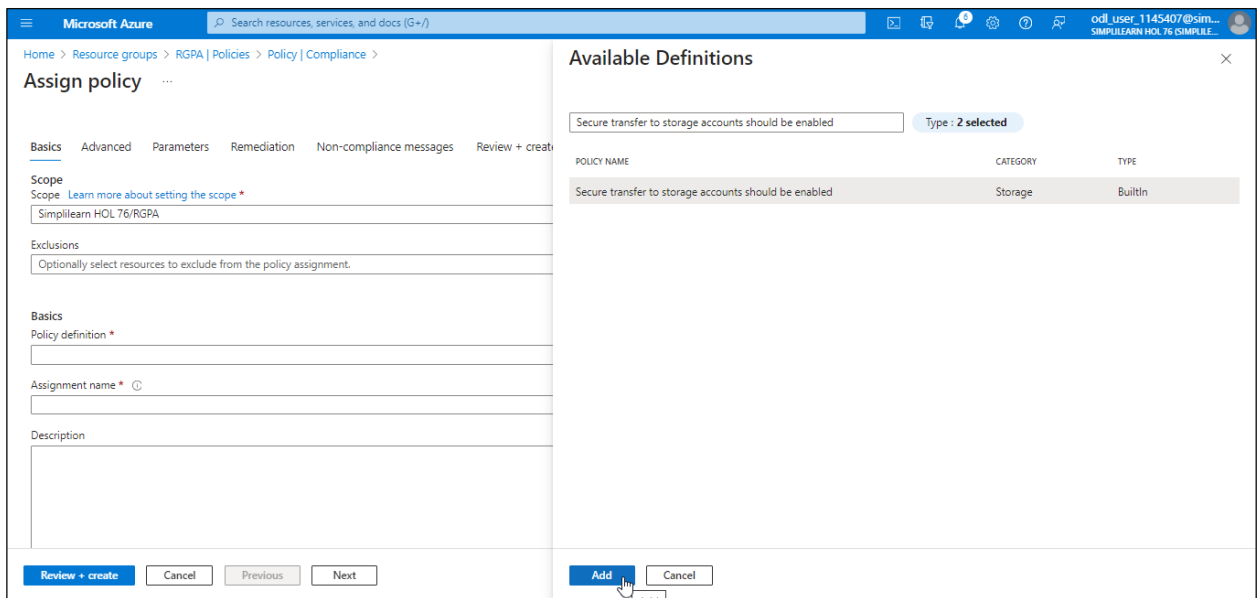
5.9 Similarly, assign the second policy by clicking on the **Assign policy** button.

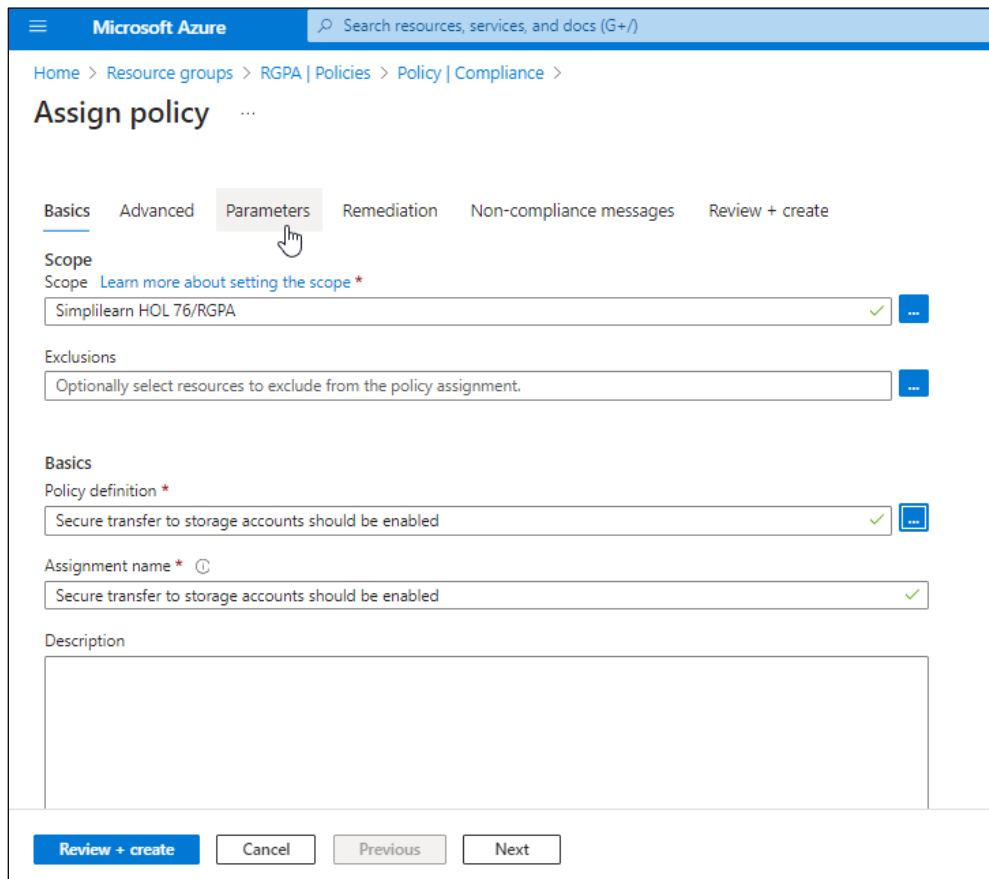


5.10 In the **Assign policy** page, click on **Policy definition** as shown in the screenshot below:



5.11 In the **Available Definitions** page, search for and select **Secure transfer to storage accounts should be enabled**, then click on **Add**



5.12 Click on **Parameters** tab

The screenshot shows the Microsoft Azure portal interface for assigning a policy. The breadcrumb navigation is: Home > Resource groups > RGPA | Policies > Policy | Compliance >. The main heading is "Assign policy" with a three-dot menu icon. Below this, there are six tabs: Basics, Advanced, Parameters, Remediation, Non-compliance messages, and Review + create. The "Parameters" tab is currently selected, and a mouse cursor is pointing at it. The "Scope" section shows a dropdown menu with the value "Simplilearn HOL 76/RGPA" and a green checkmark. The "Exclusions" section has a text input field with the placeholder "Optionally select resources to exclude from the policy assignment." and a blue button with three dots. The "Basics" section contains a "Policy definition *" dropdown with the value "Secure transfer to storage accounts should be enabled" and a green checkmark. Below this is an "Assignment name *" dropdown with the value "Secure transfer to storage accounts should be enabled" and a green checkmark. At the bottom, there is a "Description" text area. The footer contains four buttons: "Review + create" (blue), "Cancel", "Previous", and "Next".

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups > RGPA | Policies > Policy | Compliance >

Assign policy ...

Basics Advanced **Parameters** Remediation Non-compliance messages Review + create

Scope
Scope [Learn more about setting the scope *](#)

Simplilearn HOL 76/RGPA ✓ ...

Exclusions
Optionally select resources to exclude from the policy assignment. ...

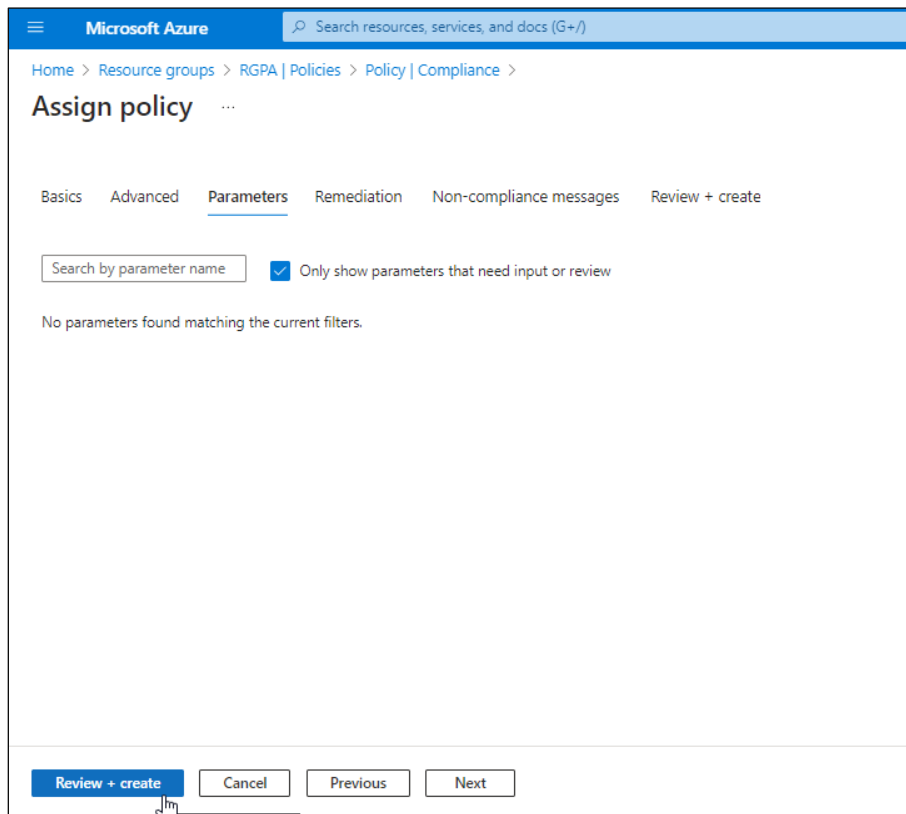
Basics
Policy definition *
Secure transfer to storage accounts should be enabled ✓ ...

Assignment name * ⓘ
Secure transfer to storage accounts should be enabled ✓

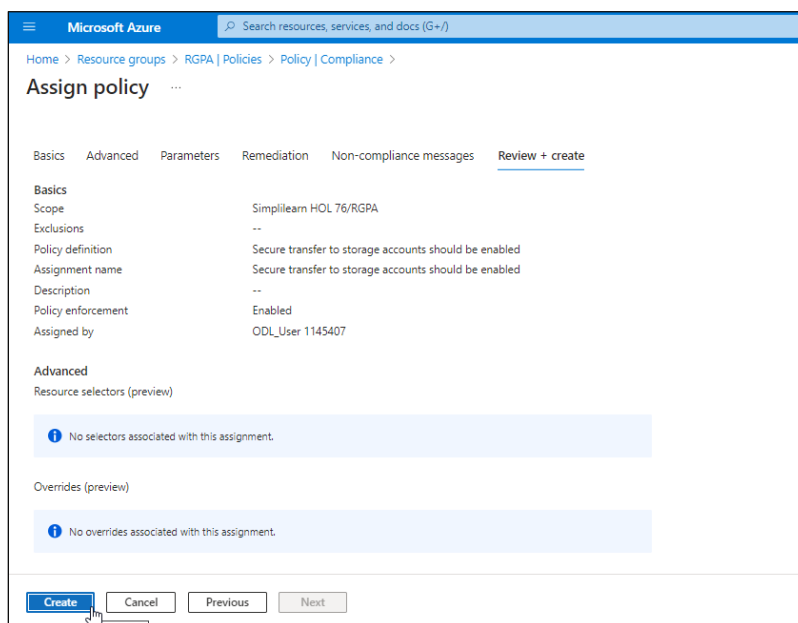
Description

Review + create Cancel Previous Next

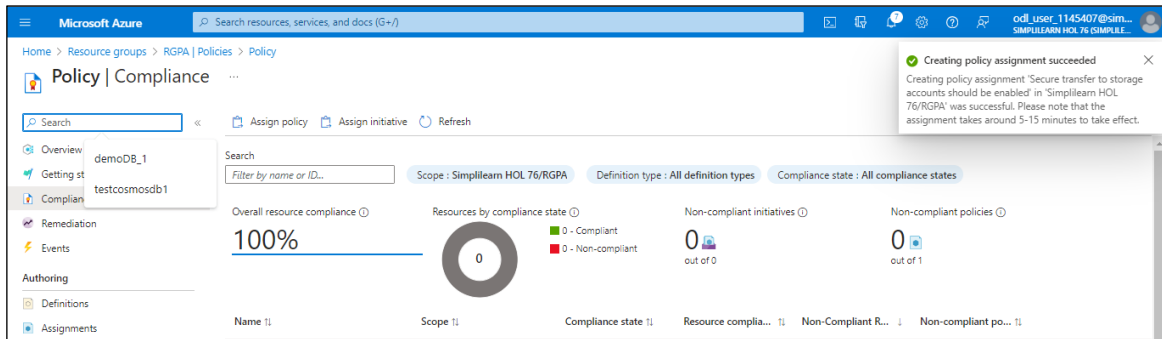
5.13 Click on **Review + create** button



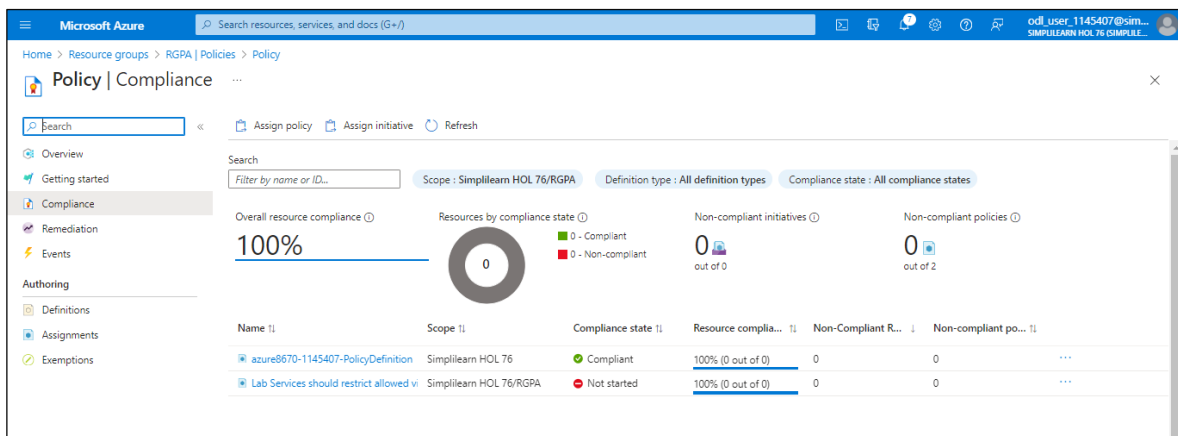
5.14 After reviewing all the details, click on **Create**



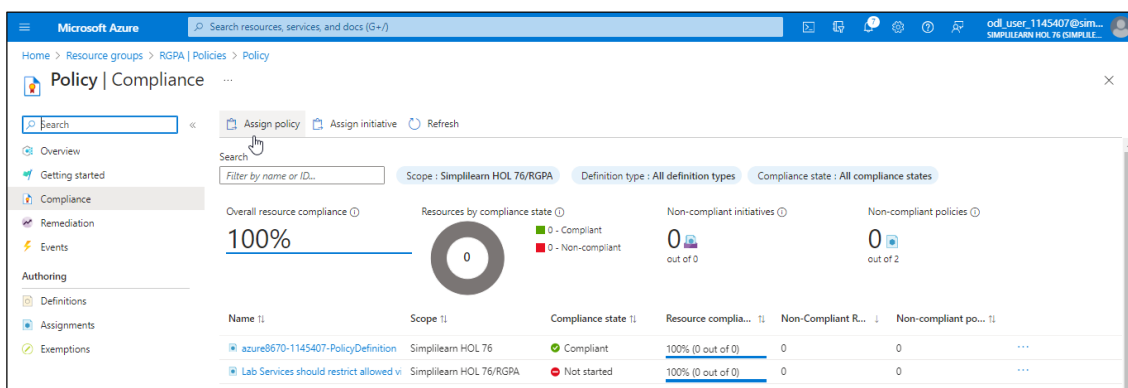
The following message will appear on the screen once the policy assignment is successful:



In the **Policy | Compliance** page, the second policy is added as shown in the screenshot below:



5.15 Now, add the third policy by clicking on **Assign policy**.



5.16 Under **Basics** field, click on **Policy definition**

Microsoft Azure | Search resources, services, and docs (G+)

Home > Resource groups > RGPA | Policies > Policy | Compliance >

Assign policy

Basics | Advanced | Parameters | Remediation | Non-compliance messages | Review + create

Scope
 Scope [Learn more about setting the scope](#)
 Simplilearn HOL 76/RGPA

Exclusions
 Optionally select resources to exclude from the policy assignment.

Basics
 Policy definition *
 Assignment name *
 Description

[Review + create](#) | [Cancel](#) | [Previous](#) | [Next](#)

5.17 In the **Available Definitions** page, search for and select **Allowed locations**, then click on **Add**

Microsoft Azure | Search resources, services, and docs (G+)

Home > Resource groups > RGPA | Policies > Policy | Compliance >

Assign policy

Basics | Advanced | Parameters | Remediation | Non-compliance messages | Review + create

Scope
 Scope [Learn more about setting the scope](#)
 Simplilearn HOL 76/RGPA

Exclusions
 Optionally select resources to exclude from the policy assignment.

Basics
 Policy definition *
 Assignment name *
 Description

[Review + create](#) | [Cancel](#) | [Previous](#) | [Next](#)

Available Definitions

Allowed locations Type: 2 selected

POLICY NAME	CATEGORY	TYPE
Azure Cosmos DB allowed locations	Cosmos DB	Builtin
Allowed locations	General	Builtin
Allowed locations for resource groups	General	Builtin

[Add](#) | [Cancel](#)

5.18 Go to the the **Parameters** tab

The screenshot shows the 'Assign policy' page in the Microsoft Azure portal. The breadcrumb navigation is 'Home > Resource groups > RGPA | Policies > Policy | Compliance >'. The page title is 'Assign policy'. There are five tabs: 'Basics', 'Advanced', 'Parameters' (which is selected and has a mouse cursor over it), 'Remediation', and 'Non-compliance messages'. Below the tabs, there are sections for 'Scope' (with a link 'Learn more about setting the scope *'), 'Exclusions', 'Policy definition *' (set to 'Allowed locations'), 'Assignment name *' (set to 'Allowed locations'), and a 'Description' text area. At the bottom, there are buttons for 'Review + create', 'Cancel', 'Previous', and 'Next'.

5.19 Add locations under the **Allowed locations** field by clicking on the drop-down and selecting the regions for the developers to operate as shown in the screenshot below:

This screenshot shows the 'Parameters' tab of the 'Assign policy' page. It includes a search bar 'Search by parameter name' and a checkbox 'Only show parameters that need input or review' which is checked. The 'Allowed locations *' section shows a dropdown menu with '2 selected'. A list of locations is displayed with checkboxes: 'Select all', 'East Asia (Stage)', 'East US' (checked), 'East US (Stage)', 'East US 2' (checked), 'East US 2 (Stage)', 'Europe', 'France', and 'France Central'. At the bottom, the same navigation buttons 'Review + create', 'Cancel', 'Previous', and 'Next' are visible.

5.20 Click on **Review + create**

The screenshot shows the 'Assign policy' page in the Microsoft Azure portal. The breadcrumb trail is 'Home > Resource groups > RGPA | Policies > Policy | Compliance >'. The page title is 'Assign policy ...'. There are five tabs: 'Basics', 'Advanced', 'Parameters' (selected), 'Remediation', and 'Non-compliance messages'. Below the tabs, there is a search box 'Search by parameter name' and a checked checkbox 'Only show parameters that need input or review'. Below that is a dropdown menu 'Allowed locations * ⓘ' showing '2 selected'. At the bottom, there are four buttons: 'Review + create' (highlighted with a mouse cursor), 'Cancel', 'Previous', and 'Next'.

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups > RGPA | Policies > Policy | Compliance >

Assign policy ...

Basics Advanced **Parameters** Remediation Non-compliance messages Review + create

Search by parameter name ☒ Only show parameters that need input or review

Allowed locations * ⓘ
2 selected

Review + create Cancel Previous Next

5.21 In the **Review + create** tab, after reviewing the details, click on **Create**

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Resource groups > RGPA | Policies > Policy | Compliance >

Assign policy ...

Basics Advanced Parameters Remediation Non-compliance messages **Review + create**

Basics

Scope	Simplilearn HOL 76/RGPA
Exclusions	--
Policy definition	Allowed locations
Assignment name	Allowed locations
Description	--
Policy enforcement	Enabled
Assigned by	ODL_User 1145407

Advanced

Resource selectors (preview)

No selectors associated with this assignment.

Overrides (preview)

No overrides associated with this assignment.

Create Cancel Previous Next

The following message will appear on the screen once the policy assignment is successful:

Microsoft Azure Search resources, services, and docs (G+/I)

Home > Resource groups > RGPA | Policies > Policy | Compliance

Policy | Compliance

Assign policy Assign initiative Refresh

Search Filter by name or ID... Scope: Simplilearn HOL 76/RGPA Definition type: All definition types Compliance state: All compliance states

Overall resource compliance **100%**

Resources by compliance state

0 - Compliant	0
0 - Non-compliant	0

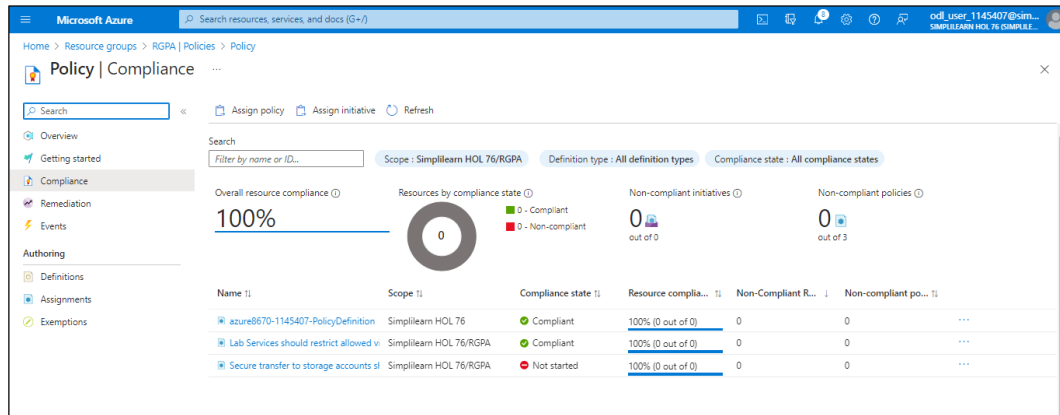
Non-compliant initiatives 0 out of 0

Non-compliant policies 0 out of 2

Creating policy assignment succeeded

Creating policy assignment 'Allowed locations' in 'Simplilearn HOL 76/RGPA' was successful. Please note that the assignment takes around 5-15 minutes to take effect.

In the **Policy | Compliance** page, the third policy is added as shown in the screenshot below:



By following the above steps, you have successfully designed a solution for developers, allowing them to provide Azure resources according to the stated requirements.